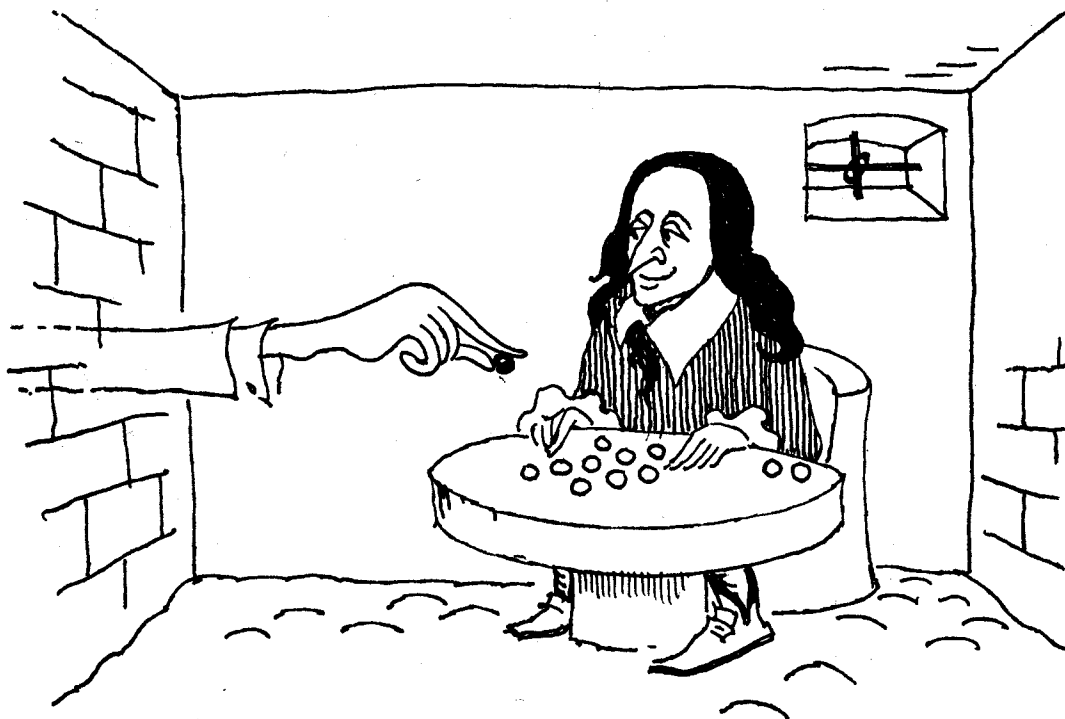


POLYTAIRE™

A
PIET HEIN
game



Why POLYTAIRE is like it is

It seems to be almost a matter of chance which games man has invented or rather discovered among the multitude of possibilities. And the games that happened to be invented have retained their popularity and become classics – regardless of whether there are better and richer games possibilities very close to them.

One of the examples is Solitaire. It has existed for centuries, and throughout this time it has been confined quite unnecessarily to a rectangular pattern, although a triangular pattern gives far greater possibilities of variation, and allows far more problems. In addition, Solitaire has been limited to a game for a single player, just as »patience« or jigsaw puzzles, although in a very obvious way and without losing any simplicity it can be expanded into a game for two players.

If one doesn't accept the traditional limitations of a game, but makes use of one's freedom to explore all the possible

variations, one often discovers entirely new, richly varied games.

This is what has happened in the case of Solitaire. It has been expanded into a more varied game for a single player, as well as a wholly new, surprisingly simple and varied game for two – the game to which I have given the name POLYTAIRE – alluding to the expansion by the Greek word *poly* = many.

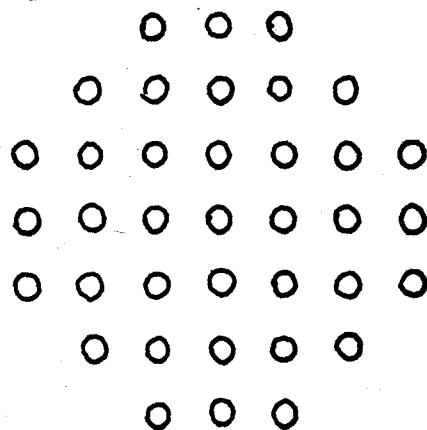
POLYTAIRE refers partly to *Solitaire*, of which, as mentioned, it is a significantly enlarged version, and partly to its salient feature which makes the game more varied by the use of the new pattern allowing moves in 6 directions, and is not just a kind of patience for one but gives possibilities of a game for two players.

POLYTAIRE illustrates the fact that the simplest things are not always the result of the simplest thinking – the path

to them is long and often leads at first through the more complicated.

The first steps were to give the board a triangular pattern instead of the rectangular. This proved an enrichment without the game losing anything in simplicity.

But when I wished to transform it into a game for two players, I fell automatically into the trap of making the aim of the game be to remove as many as possible of the opponent's balls from the board. Consequently both players made their balls as far as possible creep around the edges of the board where there were no holes behind them and thus no chance of capture. The game then got tied in a knot at an early stage with lots of uncapturable balls of both colours.



POLYTAIRE

POLYTAIRE is an expansion of the ancient solo game *Solitaire* and thus it is partly:

a more richly varied game for one player and partly:

a game for two players based on the same principles, but of such a nature that it can be played quite straightforwardly or so strategically that one thinks several moves in advance just as in chess.

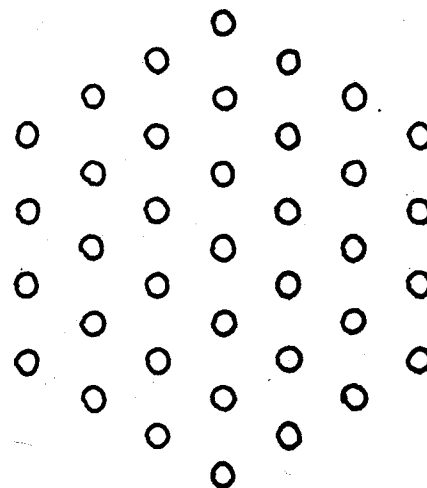
THE POLYTAIRE BOARD

is played on a board which like that of *Solitaire* has 37 squares or holes, but in contrast to *Solitaire* these are placed in a triangular pattern giving a hexagonal board.

How was I to untie the knot and entice the balls out into the open?

The solution was quite simple: let the aim of the game be to remove as many as possible of one's own balls from the board – a principle which is far simpler and more logical and should be the more obvious solution as it is more closely related to the principle in *Solitaire*.

It is worth mentioning how richly varied and free from boring interludes is the simple game which hereby evolved when compared, for example, with Chinese Chequers, especially considering how much smaller area – 37 holes as against 121.



POLYTAIRE FOR ONE

Just as in *Solitaire*, the game starts with a full board: a ball in every hole except one. Then you let each ball, one at a time, jump over a neighbouring ball into a vacant hole just behind, whereafter you remove the ball over which you have jumped. In this way all the balls excepting one can be removed from the board.

In contrast to *Solitaire*, where the game can be solved only if the hole empty at the start happens to be one of a certain 16 of the 37 holes, POLYTAIRE is supposed to be solvable regardless of which hole is empty at the start, and regardless of which hole you decide the last ball must end in. This makes it possible to pose $37 \times 37 = 1369$ problems.

In playing the game alone, the different colours of the balls are of no importance.

DOUBLE-POLYTAIRE FOR ONE

Having balls of two colours provides possibilities of yet another variation of POLYTAIRE for a single player.

Start with a completely empty board.

There are 18 white, 18 black, and one gold ball in the leather bag.

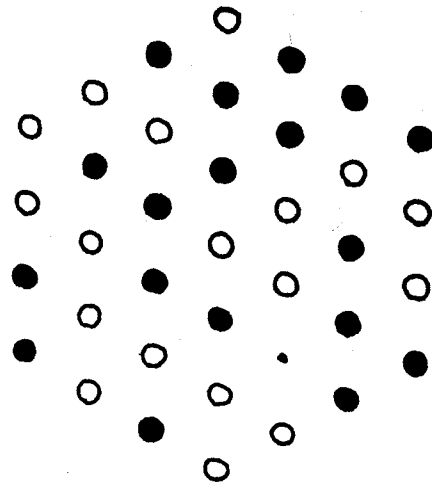
Take the balls one by one out of the bag without regard to colour, and place them in the holes on the board in this specific order: first in the upper row, the row farthest away from you, in the four holes from left to right, and then in the row beneath in the five holes from left to right, etc.

In this way the black and the white balls are placed in an utterly random order, and the gold ball in a random hole.

The gold ball is then removed from the board giving the vacant hole with which to start the game.

Then the rule is introduced that both white and black balls can jump, and that they can jump over balls of either colour, but white balls may only be removed from the board if it is white balls that jump over them, and likewise for black balls.

The objective of this game is to end up with one white and one black ball.



POLYTAIRE FOR TWO

The balls are placed as for DOUBLE-POLYTAIRE for one player.

The two players take turns to move.

White starts.

A move consists of letting one of your own balls jump over a neighbouring ball to land in a vacant hole immediately behind this: if desired you may carry on jumping from there, so long as there is an empty hole behind a neighbouring ball.

Both balls of one's own colour and of your opponent's colour may be jumped over. Balls of your own colour over which you jump are removed from the board, but not balls of your opponent's colour.

The aim of the game is to reduce the number of your own balls as much as possible.

If a player can make a move, he must do so, but he is not required to make more than one jump, even if several are possible.

If one player cannot make a move, then the other takes his turn.

When neither player can move then the game finishes, and the player with the least number of balls on the board is the winner.

Even though a player only has a single ball left – and therefore no chance of further reducing his number of balls – he must move if and when he can. His moves may be of significance, since by different positions he can prevent his opponent from reducing his score.

In the individual moves it is often important to think out several moves in advance; and good play is not merely a matter of removing as many of your own balls as possible in a move, but also of the situation in which you thus place your opponent when he has to make a move, etc. Consequently almost every move is of importance.

MINI-POLYTAIRE FOR ONE

In order to obtain an immediate impression of the problems in POLYTAIRE it is a good idea to start with a game on a smaller and simpler board, that is the board that is formed by the normal, hexagonal board with the 37 holes in a triangular pattern, when the outer row of holes is ignored; thus the play takes place on a smaller hexagonal board with 19 holes – one being vacant from the start. The other 18 are, for example, filled with the 18 black balls. Quite simply the outer row of holes is empty and remains empty throughout the game.

Also in MINI-POLYTAIRE it is supposed to be possible to impose any conditions you like as to the position of the empty hole and still solve the problem; the empty hole may have any position at the start, and the last ball may end up in any predecided hole.

In particular you can start by letting the centre hole be empty, and decide that the last ball is to end up in this hole... at all events, this has been proved possible.

Also MINI-POLYTAIRE can be played according to the rules for DOUBLE-POLYTAIRE.

HISTORY

RATHER DOUBTFUL

It has been claimed that the game of Solitaire on which POLYTAIRE is based, and which has existed for at least 300 years, was invented and played by a great French philosopher to keep himself from going mad when imprisoned in solitary confinement for gambling debts. It has even been rumored that the prisoner and inventor was the eminent French mathematician, Blaise Pascal.

Whoever it was: while other games led the inventor into debt and prison, this game helped him during the long solitary confinement – yet another proof of the game's admirable properties in relation to other games.

In comparison to the original form of Solitaire, the new form, POLYTAIRE, is much more varied in numerous ways,

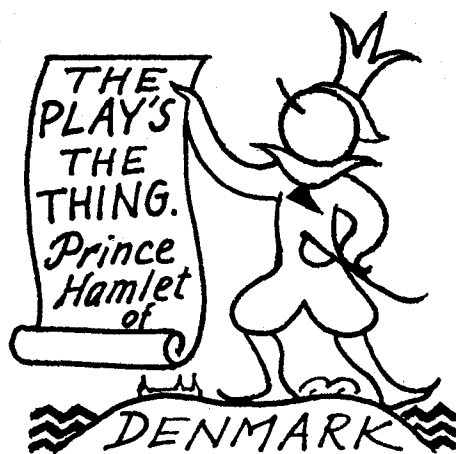
amongst others as it can be played by two players – a useful feature to-day when solitary confinement is not quite as common as it was in Pascal's time.

Pascal, who was a pioneer mathematician, wrote about his work, »No one shall claim that I have said nothing new. The field has been given a new form.«

In the same sense the conversion of his game has created something new.

And Pascal added, as a jibe to his colleague, Descartes, among others, »We both play tennis with the same ball, but one of us plays better than the other.«

The same very often applies to two people who play POLYTAIRE.



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